

Standard Operating Procedure

KT-1 / KT-2 / KT-4 / KT-400 Controller Default & Configuration Procedure

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Applies To	Field Technicians, Installers	Jurisdiction	Ontario / Manitoba
Owner	Apex – Operations	Approved By	

1. Purpose

This procedure defines the standard method for resetting Kantech KT-1, KT-2, KT-4, and KT-400 door controllers to factory default and configuring them for use on an Apex-installed access control system. It ensures consistent commissioning across all sites and reduces field errors during installation and troubleshooting.

2. Scope

Applies to all Apex technicians installing, replacing, or troubleshooting Kantech KT-1, KT-2, KT-4, and KT-400 controllers on EntraPass Special, Corporate, or Global Edition systems.

3. Regulatory & Reference Standards

- Ontario Occupational Health and Safety Act (OHSA) and O. Reg. 213/91 – Construction Projects
- Manitoba Workplace Safety and Health Act
- CSA C22.1 – Canadian Electrical Code, low-voltage wiring practices
- Manufacturer reference: Kantech KT-1 / KT-2 / KT-4 / KT-400 Installation Guides (Johnson Controls / Kantech Knowledge Exchange)

4. Tools & Materials Required

- Laptop with KT-Finder installed and EntraPass workstation access
- Ethernet patch cable
- Small flathead screwdriver (for jumper removal on KT-400)
- Site IP addressing scheme / network configuration sheet
- PPE per SOP-LV-004

5. Factory Default – KT-1 / KT-2

KT-1 and KT-2 controllers use a single Reset button on the main board — no jumpers are required.

1. Power down or isolate the controller if instructed by site procedure; otherwise the controller may remain powered.
2. Remove the front cover by unlocking the clip at the bottom.
3. Locate the Reset button in the lower right of the main board.
4. To perform a Factory Default DHCP reset: press the Reset button three (3) times, then hold for 3 seconds until you hear 5 beeps.

5. Confirm the Kantech shield LED shows the Factory Default DHCP pattern (continuous long pulses).
6. Replace the front cover.

6. Factory Default – KT-4

KT-4 controllers use a single Reset button on the main board — no jumpers are required. The button sequence confirms with a hold-and-beep step before and after the taps.

1. Locate the Reset button in the upper left of the main board.
2. Press and hold Reset for 3 seconds until you hear 5 beeps.
3. Press Reset three (3) times (for Factory Default DHCP).
4. Press and hold Reset for 3 seconds again until you hear 5 beeps to confirm.
5. Confirm the reset feedback LED and buzzer indicate the factory default DHCP pattern.

7. Factory Default – KT-400

KT-400 controllers use jumpers JP2/JP3 in combination with the Reset button.

1. Before resetting, confirm: Ethernet is connected to the network (green LINK LED lit, yellow TXRX LED flashing) and the blue Heartbeat LED is flashing.
2. Remove jumpers JP2 and JP3.
3. Press the Reset button.
4. Check the blue Heartbeat LED pattern to confirm factory default DHCP mode.
5. Verify the controller's IP address using KT-Finder.
6. Reinstall jumpers JP2 and JP3.
7. Proceed to configuration using KT-Finder.

8. Configuration (All Controllers)

1. Connect the laptop to the same network/subnet as the controller(s) and open KT-Finder.
2. Confirm each new controller appears in the list, identified by MAC address.
3. Assign a static IP address per the site's IP addressing scheme (recommended for EntraPass-managed controllers).
4. Set the operating mode to EntraPass mode (not Standalone/Cloud) unless the site specifically requires standalone operation.
5. In EntraPass, go to Devices → Connection configuration and add/confirm the connection.
6. In EntraPass, go to Devices → Configuring controllers, add the controller, select the correct model (KT-1, KT-2, KT-4, or KT-400), and enter its IP/MAC address.
7. Reload the controller (Operations → Manual operations on controllers → Reload) to push door, input, and relay configuration to the unit.

9. Verification

- Controller shows Online status in EntraPass
- Test at least one door, one input, and one relay per controller before terminating remaining field wiring
- Confirm event messages appear correctly in the EntraPass message desktop
- Record controller MAC address, assigned IP, and location on the site network configuration sheet

10. Notes

If a controller fails to appear in KT-Finder after a factory default reset, confirm the laptop and controller are on the same broadcast domain/subnet, and that no managed switch port security is blocking DHCP broadcast traffic.